Multi-Source Data Fusion for Explainable Corporate Financial Distress Prediction

[Extended abstract]

Loris Cino[1,*], Stefano Piersanti[2] and Aris Anagnostopoulos[1]

[1]Department of Computer, Control, and Management Engineering (DIAG), Sapienza University of Rome, Via Ariosto 25, 00185 Rome, Italy
[2]Bank of Italy, Via Nazionale 91, 00184 Rome, Italy

Abstract

Financial institutions and supervisors must prioritize firms for monitoring from large, heterogeneous collections of corporate records. We study one-year-ahead risk prediction for two related but distinct outcomes: legal bankruptcy and banking-system default. Our ongoing work integrates raw balance sheets, financial ratios, two complementary sources of granular credit information, and company attributes for Italian non-financial firms. The resulting corpus contains 5,480,621 firm-year observations for 473,957 firms. In an out-of-time evaluation, accounting information is the strongest individual source for bankruptcy, whereas credit behavior is more informative for default. Nevertheless, fusing heterogeneous sources yields the best overall predictive performance for both tasks. CatBoost reaches ROC-AUC values of 0.918 and 0.947 for bankruptcy and default, respectively. SHAP explanations reveal correspondingly different evidence: balance-sheet strength and company context dominate bankruptcy predictions. Credit utilization, interest rates, and arrears dominate default predictions. These results show how source-aware fusion and task-specific explanations can support transparent risk-based prioritization, also exposing the limits of threshold-independent evaluation under severe class imbalance.

Keywords

risk prediction, multi-source data fusion, financial distress, explainable machine learning, imbalanced evaluation

1. Introduction

Risk-assessment systems face an information-access problem: given a large collection of firms that ask for credit, which entities should an analyst inspect more before deciding to grant or not a loan? A predictive score provides a risk-based ordering that can reduce this collection to a manageable watchlist. Although our models are trained for binary classification rather than with a learning-to-rank objective, the problem can be viewed operationally as ranking: continuous distress scores order firms by estimated risk, allowing analysts to prioritize a limited inspection capacity. Unlike document retrieval, however, relevance here is a future event and entities are represented by structured evidence from several institutional sources. Our primary methodological focus therefore remains financial-distress prediction, data fusion, and evaluation; ranking is a downstream use of the resulting scores.

Corporate distress research often treats *bankruptcy* and *default* as interchangeable, although they encode different events. Bankruptcy is a formal legal status. The Bank of Italy's adjusted default captures distress observed across the banking system, including severe past-due and unlikely-to-pay positions. Accounting models have a long history in bankruptcy prediction [1, 2]; credit-account activity can instead provide early signals of default [3], and prior work has demonstrated the value of machine learning on Italian credit and accounting data [4]. What remains unclear is whether the same information sources and explanations support both prediction tasks when the targets, population, and evaluation protocol are held fixed.

We address this gap through a unified comparison with three contributions. First, we evaluate bankruptcy and default on the same institution-scale corpus and temporal split. Second, we quan-

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*Corresponding author.
cino@diag.uniroma1.it (L. Cino)
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tify how accounting, credit, and contextual sources complement one another. Third, we explain the resulting predictions with SHAP, showing that the source-level performance differences correspond to different task-specific drivers.

2. Data and Experimental Setup

Tasks and sources. Each record represents a non-financial Italian firm in a given year; features at year t are used to predict bankruptcy or adjusted default at $t + 1$. The corpus spans 2010–2024 and contains 5,480,621 firm-year observations from 473,957 firms. It includes 72,724 bankruptcy events and 157,085 default events, making bankruptcy both rarer and empirically harder to identify.

We organize predictors into five groups. *B* contains raw balance-sheet items; *R* contains established ratios derived from those accounts [1, 2]; *C* payment behavior from the Italian Central Credit Register; *A* contains loan-level AnaCredit information available since 2018; and *Att* contains sector, location, size, and firm-age attributes. Monthly or quarterly loan records are aggregated to firm-year statistics. Credit amounts are also normalized by total granted credit, producing behavioral ratios that remain comparable across firm sizes. Firms without reported credit are retained: missing credit amounts are set to zero and other missing fields remain marked as unavailable.

Protocol. To prevent temporal leakage and have a fair comparison, all reported experiments train on 2018–2023 and test on 2024. Missing values are handled without deleting firm-years, because missingness itself can be associated with distress. The training sample is under-sampled to a 10:1 ratio of healthy to distressed firms. We train each configuration with five random seeds and report the mean and standard deviation of each metric. CatBoost [5] for the SHAP explanation using tree-based SHAP explanations.

We report F1, Matthews correlation coefficient (MCC), ROC-AUC, and PR-AUC. MCC uses all cells of the confusion matrix and is informative under imbalance [6]; PR-AUC emphasizes the rare positive class [7]. ROC-AUC measures discrimination across thresholds and can be interpreted as the probability that a distressed firm receives a higher score than a healthy one. A high value, however, need not imply an operationally effective threshold. Accordingly, the results below distinguish score-based discrimination from thresholded classification.

3. Results: Complementary Sources

Table 1 reports representative source configurations. Scores are shown on a 0–100 scale. *All(C)* combines B, R, C, and Att; *All* additionally includes A, including the bank-provided probability-of-default (PD) field. The comparison with All(C) therefore makes the incremental role of AnaCredit and PD explicit instead of treating PD as an ordinary input.

Table 1
CatBoost performance (mean ± standard deviation over five seeds) for representative data configurations. Best mean values for each task are in bold.

	Bankruptcy			Default		
Data	F1	ROC-AUC	PR-AUC	F1	ROC-AUC	PR-AUC
B	21.75 ± 0.30	90.55 ± 0.07	13.24 ± 0.16	29.32 ± 0.22	89.32 ± 0.05	24.98 ± 0.17
C	11.71 ± 0.29	78.74 ± 0.03	8.00 ± 0.08	30.86 ± 0.31	88.68 ± 0.03	26.55 ± 0.28
All(C)	26.21 ± 0.32	91.75 ± 0.03	17.61 ± 0.34	40.92 ± 0.29	94.21 ± 0.03	38.54 ± 0.20
All	**26.29** ± 0.24	**91.81** ± 0.03	**17.97** ± 0.08	**41.46** ± 0.22	**94.68** ± 0.04	**40.05** ± 0.12

The ablation reveals different predictive signals for the two outcomes. Balance sheets alone provide strong bankruptcy discrimination (ROC-AUC 90.55), whereas C alone is substantially weaker (78.74).

For default, C improves F1 and PR-AUC over B despite a similar ROC-AUC. Adding company context and derived ratios to accounting and Central Credit Register data produces a large gain for both tasks. The complete fusion gives the highest scores, although its improvement over All(C) is modest for bankruptcy and larger for default. This asymmetry is consistent with AnaCredit and PD being designed around banking-system distress.

A wider benchmark confirms that CatBoost is competitive with an AutoML ensemble: AutoGluon [8] obtains ROC-AUC 91.92 ± 0.04 and 94.64 ± 0.06 on bankruptcy and default. We retain CatBoost for explanation because the small accuracy difference does not justify replacing a directly interpretable tree ensemble with a considerably more complex stack. More importantly, bankruptcy ROC-AUC above 0.91 coexists with PR-AUC below 0.18. This gap cautions against presenting discrimination alone as evidence of deployment readiness.

4. Task-Specific Explanations

We compute SHAP values [9] on the 2024 test set for CatBoost trained on all sources *without* the bank-provided PD, so the explanations describe signals learned from the underlying records rather than reproducing an existing risk score. In Figures 1 and 2, positive SHAP values increase predicted distress; color encodes the feature value, and prefixes identify the source (B, R, C, A, or Att).

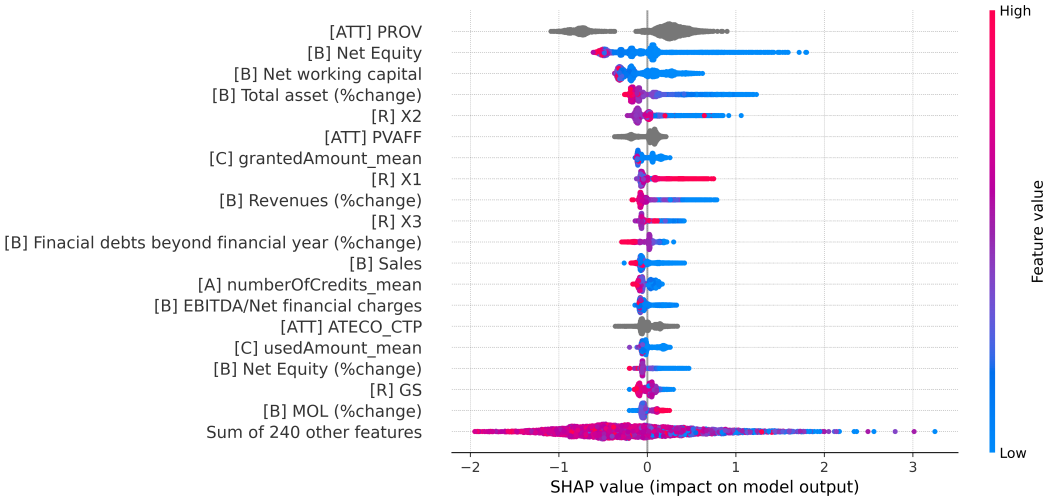


Figure 1: SHAP summary for bankruptcy prediction. Accounting strength and company context dominate the highest-ranked features.

For bankruptcy, location and sector attributes interact with measures of firm strength. Net equity and net working capital are among the most influential raw accounting variables. X1 (working capital over total assets) and X2 (retained earnings over total assets) connect the learned model to classical distress analysis. Low net equity and working capital generally move predictions toward bankruptcy. The prominence of contextual attributes does not imply a causal geographic or sector effect; rather, it indicates that the interpretation of accounting values is population-dependent.

The default explanation is qualitatively different. The most influential feature is used credit divided by granted credit, followed by the AnaCredit interest rate. Credit margin variability, normalized overdraft, the number of credits, and arrears also rank highly. Larger utilization and overdraft ratios tend to increase predicted risk, agreeing with evidence that account activity anticipates borrower default [3]. Relative credit measures are more prominent than absolute amounts, suggesting that behavior with respect to available credit is more transferable across firms of different sizes.

Taken together, the explanations provide a consistency check on the fusion experiment: the source that works best in isolation also supplies many of the most influential features for its corresponding task. At the same time, both plots contain signals from multiple sources, explaining why fusion out-

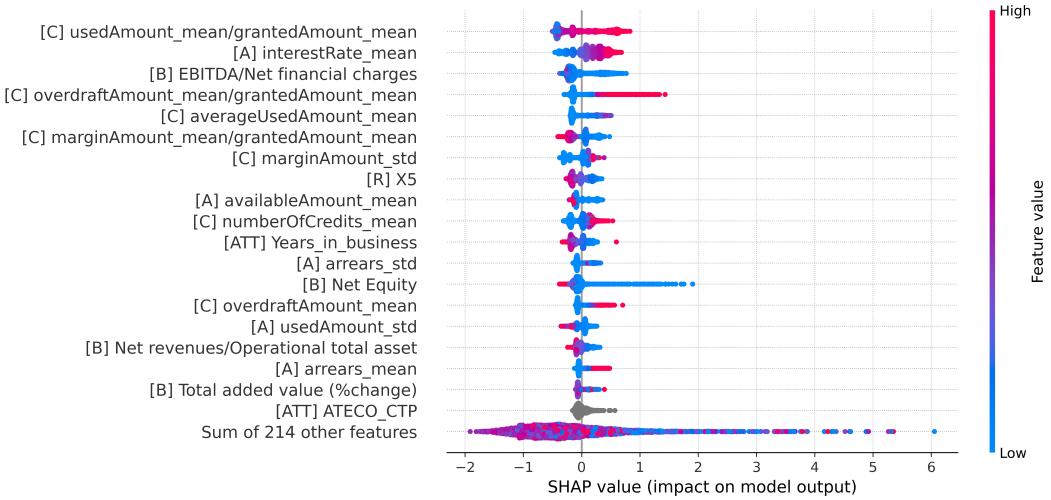


Figure 2: SHAP summary for default prediction. Credit utilization, loan price, and repayment-behavior variables provide the strongest evidence.

performs single-source models. In an analyst-facing decision-support system, these explanations could accompany each prioritized firm as evidence for why it entered a watchlist.

5. Conclusion

Our results frame corporate monitoring as explainable financial-distress prediction over heterogeneous structured records. Accounting information is most useful for predicting legal bankruptcy, credit behavior is most useful for banking-system default, and their fusion improves both tasks. SHAP analysis shows that this difference is substantive rather than merely numerical: the two predictions depend on distinct, economically plausible evidence.

The study is limited to Italian firms and proprietary institutional data, and its observational explanations should not be interpreted causally. Results are also sensitive to time: older training observations can degrade performance, fixed decision thresholds may not match the inspection capacity or error costs of a real institution. Future work will therefore study time-decayed training, calibrated top-k evaluation, and utility-aware ranking metrics. These extensions would connect predictive accuracy more directly to the human process of selecting a finite set of firms for investigation.

Declaration on Generative AI

During the preparation of this work, the authors used Cursor (GPT-5.6 Sol) for text drafting and language editing. The authors reviewed and edited the content and take full responsibility for the publication's content.

References

[1] E. I. Altman, Financial ratios, discriminant analysis and the prediction of corporate bankruptcy, The Journal of Finance 23 (1968) 589–609. doi:10.1111/j.1540-6261.1968.tb00843.x.

[2] F. Barboza, H. Kimura, E. I. Altman, Machine learning models and bankruptcy prediction, Expert Systems with Applications 83 (2017) 405–417. doi:10.1016/j.eswa.2017.04.006.

[3] L. Norden, M. Weber, Credit line usage, checking account activity, and default risk of bank borrowers, The Review of Financial Studies 23 (2010) 3665–3699. doi:10.1093/rfs/hhq061.

[4] M. Moscatelli, F. Parlapiano, S. Narizzano, G. Viggiano, Corporate default forecasting with machine learning, Expert Systems with Applications 161 (2020) 113567. doi:10.1016/j.eswa.2020.113567.

[5] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin, CatBoost: Unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems, volume 31, 2018, pp. 6638–6648.

[6] D. Chicco, G. Jurman, The advantages of the matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation, BMC Genomics 21 (2020) 6. doi:10.1186/s12864-019-6413-7.

[7] P. Flach, M. Kull, Precision-recall-gain curves: PR analysis done right, in: Advances in Neural Information Processing Systems, volume 28, 2015.

[8] N. Erickson, J. Mueller, A. Shirkov, H. Zhang, P. Larroy, M. Li, A. Smola, AutoGluon-Tabular: Robust and accurate AutoML for structured data, arXiv preprint arXiv:2003.06505 (2020). doi:10.48550/arXiv.2003.06505.

[9] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems, volume 30, 2017, pp. 4765–4774.